Visualizing Key Predictors of MLB Team Wins

Salary vs. Performance: What Actually Drives Wins?

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SUMMARY

We built a season-level MLB win prediction project to identify which team factors matter most.

Instead of predicting single games, we modeled total season wins using two complementary regression models.

WHY THIS MATTERS

Front offices, analysts, fans, and bettors all care about team wins.

More wins increase postseason chances, improve roster decisions, and create betting and forecasting value.

Most prior work focuses on game-level predictions, not full-season wins.

DATA & METHOD

Source: Lahman Baseball Database
Years: 2010 – 2016

Features: Payroll, ERA, Batting Avg, Home Runs, Errors, All-Star Count

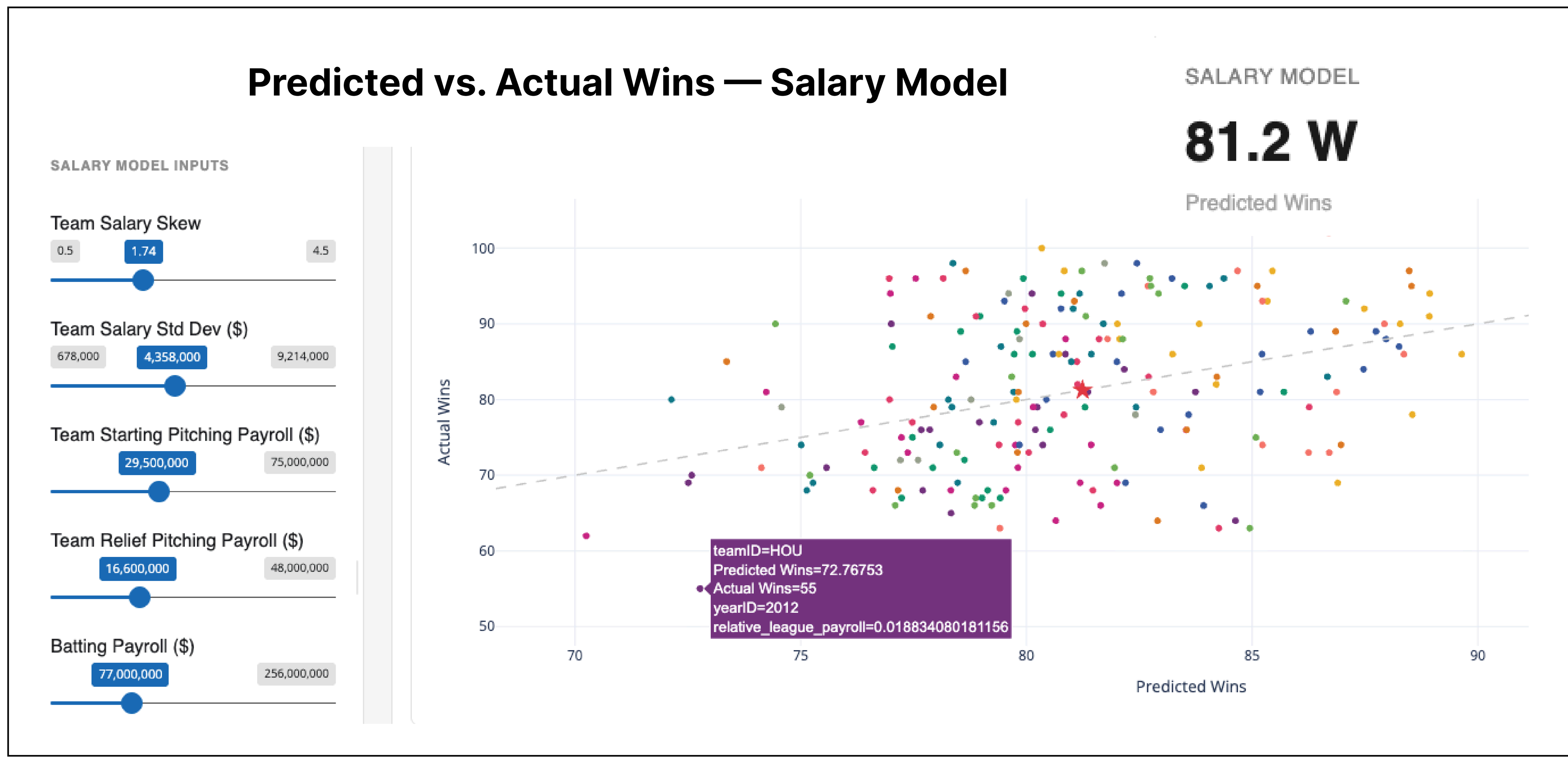
Method: Built two linear regression models

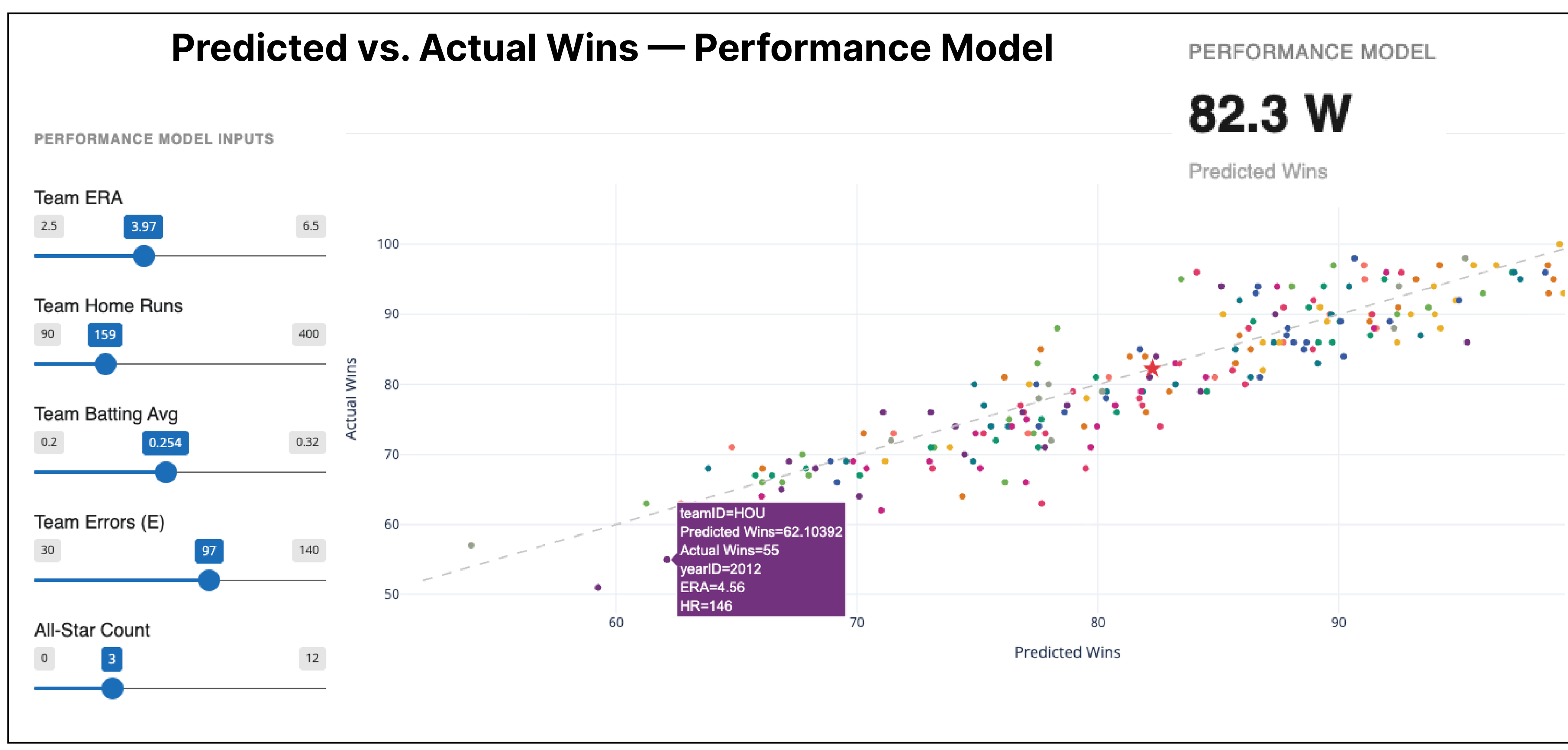
1. Salary Model – how money is distributed
2. Performance Model – how teams perform

WHAT MAKES IT DIFFERENT

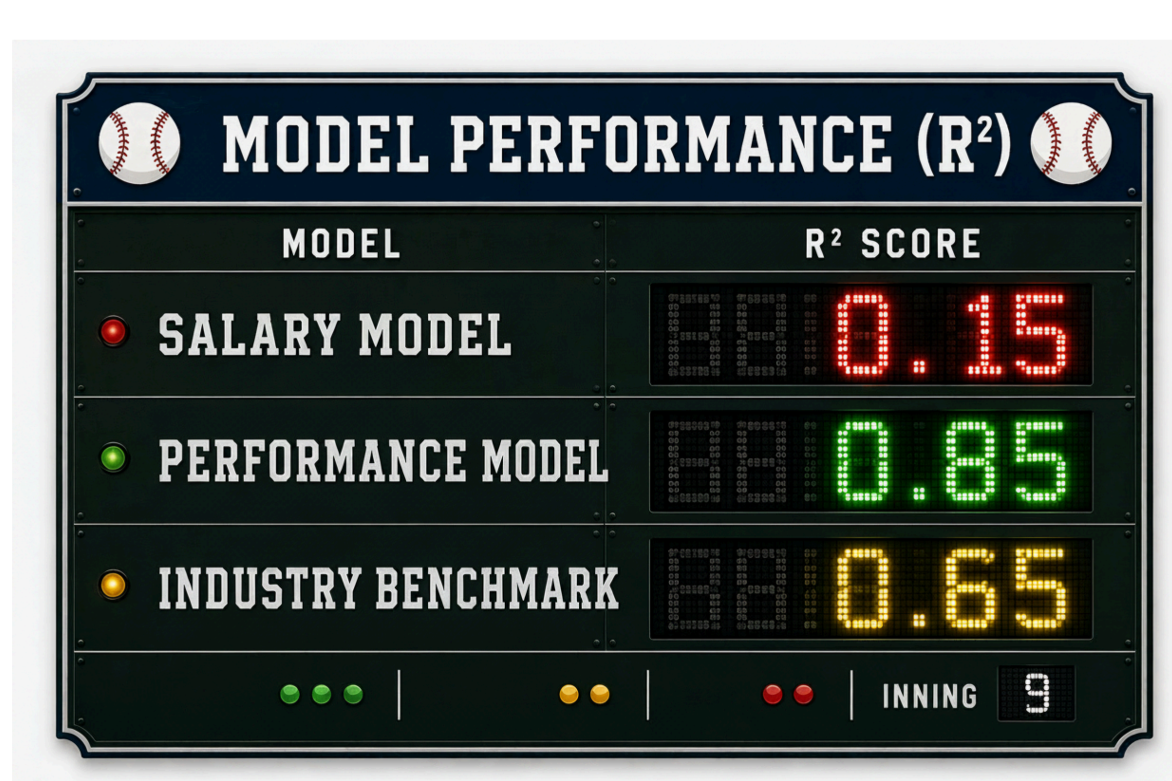
- Models total season wins (not individual games)
- Combines financial (salary) and performance metrics
- Uses modern data (2010–2016)
- Includes an interactive Shiny dashboard for exploration and decision-making

INTERACTIVE DASHBOARD PREVIEW





MODEL METRICS



Salary Model			Performance Model		
Feature	Weight	p-Value	Feature	Weight	p-Value
intercept	79.573	2E-16	intercept	41.85	1.70E-06
salary_skew	-2.715	0.1288	all_star_count	0.532	0.0515
salary_std	-0.000000583	0.5841	ERA	-17.831	< 2e-16
reliever_pitcher_payroll	1.45E-08	0.9067	HR	0.1086	< 2e-16
starter_pitcher_payroll	9.869E-08	0.1642	team_batting_avg	374.09	< 2e-16
batting_payrolll	7.51E-08	0.0896	E	-0.0403	0.083

CONCLUSION & FINDINGS

- Team salary explains only ~15% of the variation in season win totals, implying that higher purchasing power is associated with more wins, but is not highly predictive.
- Pitching ERA had the strongest negative coefficient of the performance model indicating that run prevention is the most important factor in determining wins.
- A team that improves a struggling rotation, lowering ERA by 1.00, gains 17.8 wins over the course of a season, while in comparison, the same team would need to raise its batting average by a nearly impossible 0.049 to see a similar 17 win improvement.